Decoding Biomedical Time Series into Natural Language

Using Hybrid Quantum-Classical Transformers

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Agenda Overview

01 Project Context & Motivation

02 State of the Art Review

03 Research Problem & Gaps

04 Discussion Questions

Why This Project Matters ?

Current Situation:

- Biomedical signals (EEG, ECG) contain rich health information
- Clinical interpretation and explication requires specialized expertise
- Time-consuming manual analysis limits scalability
- AI models produce numerical outputs, not explanations

State of the Art - Classical Approaches

1. Recurrent Neural Networks (LSTM/GRU)

Strengths:

- Capture temporal dependencies effectively
- Proven track record in sequence modeling

Limitations:

- Vanishing gradient problem in long sequences
- Sequential processing (slow training)
- Limited interpretability
- Struggle with non-stationarity

Reference: Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. Neural Computation, 9(8), 1735-1780.

State of the Art - Classical Approaches

2.Transformer Architectures

Strengths:

- Parallel processing with attention mechanisms
- Better scalability for long sequences

Limitations:

- Quadratic complexity $O(n^2)$
- High computational cost
- Still struggle with biomedical signal noise

Reference: Vaswani, A., et al. (2017). Attention is all you need. Advances in Neural Information Processing Systems, 30..

State of the Art - Classical Approaches

Why Classical Methods Fall Short ?

Challenge	Impact	Classical Limitation
High Noise	Signal corruption	Poor noise robustness
Non-stationarity	Shifting patterns	Assumes stationarity
High Dimensionality	Curse of dimensionality	Exponential complexity
Long-range Dependencies	Temporal context	Vanishing gradients
Interpretability	Clinical trust	Black-box models

"Classical approaches struggle to manage the inherent complexity of biomedical signals, limiting their accuracy and generalization"

State of the Art - Quantum Machine Learning

Quantum Advantage for Time Series

- Faster convergence in optimization
- Compact representations of complex patterns
- Better noise handling (quantum error correction)
- Exponential Hilbert space for feature encoding

Reference: Behera, B. K., Al-Kuwari, S., & Farouk, A. (2025). QSVM-QNN: Quantum Support Vector Machine Based Quantum Neural Network Learning Algorithm for Brain-Computer Interfacing Systems. IEEE Transactions on Artificial Intelligence, 7(1), 308-320. https://doi.org/10.1109/TAI.2025.3572852.

State of the Art - Quantum Machine Learning

1.Quantum Long Short-Term Memory (QLSTM)

- Replaces classical LSTM gates with parameterized quantum circuits (PQC)
- Quantum states encode temporal dependencies
- Variational quantum circuits for learning

Reference: Chen, S. Y., et al. (2024). Quantum long short-term memory (QLSTM) vs. classical LSTM in time series forecasting. Frontiers in Physics, 12.

State of the Art - Quantum Machine Learning

2.Quantum Transformers & Hybrid Models

Quantum Transformers:

- Variational quantum circuits for attention mechanisms
- Applications in multivariate forecasting

Hybrid Quantum-Classical CNNs (HQCNN):

- Classical backbone + quantum feature extraction
- Improved medical image classification

References: Li, G., et al. (2024). Quantum-enhanced dual-layer graph attention network for timeseries forecasting. Scientific Reports, 14(1).
Withdrawn (2025). HQCNN: A Hybrid Quantum-Classical Neural Network. arXiv:2509.14277.

State of the Art - Quantum Machine Learning

3.Quantum Long-Term Time Series Forecasting (QuLTSF)

- Outperforms classical linear models
- Reduced MSE and MAE on meteorological data

Reference: Steckhan, L., et al. (2025). QuLTSF: Long-Term Time Series Forecasting with Quantum Machine Learning. arXiv:2412.13769.

Large Language Models for Biomedical Signals

Why LLMs?

- Generate human-readable explanations
- Bridge the gap between numerical predictions and clinical understanding
- Enable natural human-AI interaction

Large Language Models for Biomedical Signals

1.LLMs for EEG - Comprehensive Survey:

- Large Language Models (LLMs) are adapted for EEG representation learning
- Can map EEG features to semantic or language-based outputs
- Support multimodal generation (EEG + text) and clinical decision-making

2.BioSignal Copilot:

- LLMs generate clinical reports from biological signals automatically
- Closest to the goal of our PFE project: bridging EEG signals and human-readable outputs

Reference: Wang, H., et al. (2023). BioSignal Copilot: Leveraging LLMs in drafting reports for biosignals. medRxiv

Hybrid Systems - The Gap

Current State

- Quantum ML shows promise for temporal data
- LLMs excel at interpretable text generation
- Hybrid systems emerging in medical imaging

Hybrid Systems - The Gap

What's Missing in Current Research?

- **Gap 1:** Quantum-Classical Hybrid Optimization
- **Gap 2:** Quantum-LLM Alignment
- **Gap 3:** Clinical Interpretability Validation

Let's discuss how to make this vision a reality!

Thank You

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